

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0224

Roz 100 kV
November 27, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0224. The Interconnection Customer proposes a 19.4 MW generating facility interconnecting at the Roz 100 kV substation in the Indiana area. The request was received on April 07, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 151 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$492,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Butler Municipal Power Plant / Hansel 100.0 kV	E Methuen 500-230 kV Autotransformer	111.5	128	5.4
Butler Municipal Power Plant / Savona 100.0 kV	East Methuen Substation 500 kV/230 kV Transformer	108.1	121	5.4
Savona / Richard Lake 100.0 kV	E Methuen 500/230 kV XFMR	107.2	110	5.4
Thousand Springs / Hansel 100.0 kV Line	E Methuen 500-230 kV Transformer	106.0	142	5.4
Gray'S Flat / Baraga Vlg 230.0 kV	Madeline / E Methuen 230.0 kV	102.0	673	13.1
15Th Avenue South / Norwich Substation 230.0 kV	Omni-Hackberry / Norwich 500.0 kV	100.7	507	6.6
Villa Park / Wolfe 230.0 kV	Wolfe / Barrett Substation 500.0 kV	100.5	703	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hansel / Butler Municipal Power Plant 100.0 kV L	Rebuild line with higher-capacity conductor	\$6,547,000	3% (\$197,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Butler Municipal Power Plant / Savona 100.0 kV	Reconductor line	\$631,000	3% (\$19,000)
Richard Lake / Savona 100.0 kV	Reconductor line	\$3,359,000	3% (\$101,000)
Hansel / Thousand Springs 100.0 kV	Reconductor line	\$1,921,000	3% (\$58,000)
Baraga Vlg / Gray'S Flat 230.0 kV	Reconductor line	\$5,280,000	1% (\$40,000)
Norwich / 15Th Avenue S 230.0 kV	Reconductor line	\$4,082,000	1% (\$30,000)
Villa Park / Wolfe 230.0 kV	Reconductor line	\$5,360,000	1% (\$47,000)
	Total allocated to project		\$492,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.